

PRAGATI BHARATI NATIONAL TALENT EXAMINATION 2026
SECTION A: COMPUTER SCIENCE & LOGICAL REASONING

1. What is the time complexity of searching an element in a balanced binary search tree?
A. $O(1)$
B. $O(n)$
C. $O(\log n)$
D. $O(n \log n)$

2. Which protocol is primarily responsible for reliable connection-oriented transfer on the internet?
A. UDP
B. TCP
C. IP
D. ICMP

3. Consider a graph $G = (V, E)$. A spanning tree of G is a subgraph that includes all vertices.
Question continues on the next page...

SECTION A (CONTINUED FROM PAGE 1)

3. Which greedy algorithm finds the Minimum Spanning Tree of a connected, undirected graph?

- A. Dijkstra's Algorithm
- B. Kruskal's Algorithm
- C. Floyd-Warshall Algorithm
- D. Bellman-Ford Algorithm

4. In SQL, which clause is used to filter records after aggregation with GROUP BY?

- A. WHERE
- B. HAVING
- C. FILTER
- D. ORDER BY

5. Which of the following is an example of an NP-Complete problem?

- A. Shortest Path in DAG
- B. Traveling Salesperson Problem
- C. Minimum Spanning Tree
- D. Merge Sort